4 Compounds **P** and **Q** are structural isomers.



Fig. 4.1

(a) (i) Define structural isomerism.

.....

..... [2]

(iii) Both **P** and **Q** react with aqueous bromine.

Name the mechanism of this reaction.

..... [1]

(iv) **Q** is oxidised by hot concentrated acidified $KMnO_4(aq)$, forming two different organic products.

Construct an equation for this reaction. Use [O] to represent an atom of oxygen from the oxidising agent.

$(CH_3)_2C{=}CHCH_3$ ..... [2]

(v) **Q** reacts with HBr(g) to produce two structural isomers, **R** and **S**, as shown in Fig. 4.2.

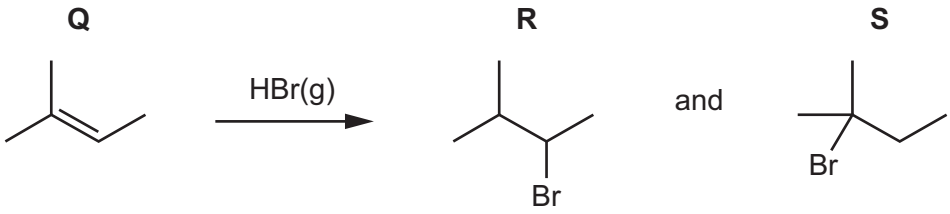


Fig. 4.2

State and explain why isomer **S** is the major product of the reaction.

.....

.....

..... [2]

3 Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

(a) Define empirical formula.

.....

..... [1]

(c) Cyclopentane reacts with Cl_2 in the presence of ultraviolet light to form C_5H_9Cl.

(i) The reaction is initiated by the bond fission of Cl_2.

State the type of bond fission shown in the initiation step.

..... [1]

(ii) Complete the equations to show the two propagation steps that follow the initiation step.

propagation 1 C_5H_{10} + ... → C_5H_9• + ...

propagation 2 C_5H_9• + ... → .....

[2]

(iii) The final step is shown.

$$C_5H_9\bullet + Cl\bullet \rightarrow C_5H_9Cl$$

Give the name for this step in the reaction.

..... [1]

(d) Fig. 3.2 shows a reaction cycle involving cyclopentane, cyclopentene and C_5H_9Cl.

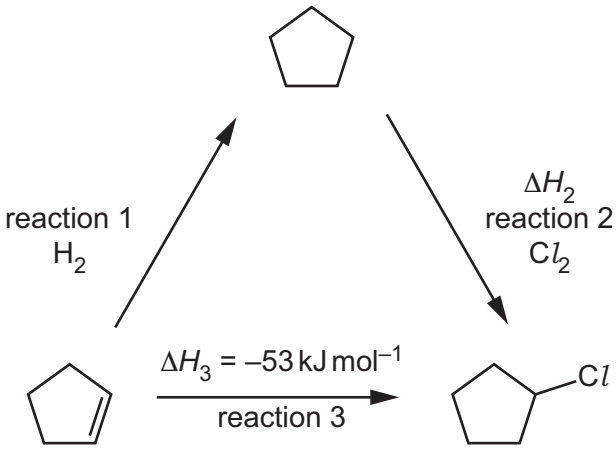


Fig. 3.2

(i) Identify a suitable reagent for reaction 3.

..... [1]

(e) Cyclopentene, C_5H_8, reacts with hot concentrated acidified $KMnO_4$ to form compound **W**, $C_5H_8O_4$.

(i) Draw the structure of **W**.

[1]

(ii) The infrared spectrum of **W** is shown in Fig. 3.3.

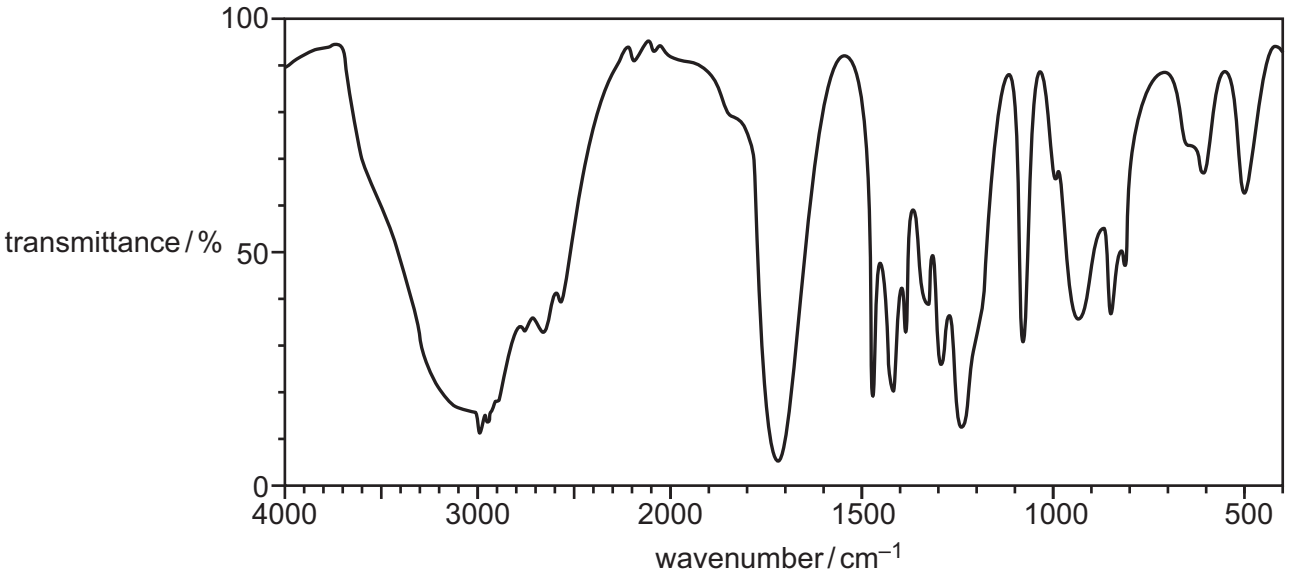


Fig. 3.3

Identify **two** absorptions in the infrared spectrum of **W** that would **not** be present in the infrared spectrum of cyclopentene.

- Write **1** or **2** on Fig. 3.3 against each of these two absorptions.
- Complete Table 3.2 to show which bond is responsible for each absorption that you have identified in Fig. 3.3.

Table 3.2

absorption	**1**	**2**
bond responsible		

[2]

Table 3.3

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers) / cm^{-1}
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

[Total: 13]

4 Fig. 4.1 shows a possible synthesis of propene, C_3H_6.

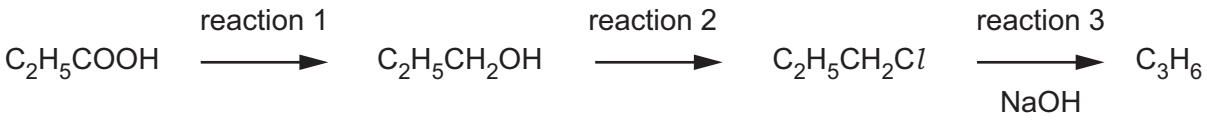


Fig. 4.1

(a) (i) Identify the type of reaction that occurs in reaction 1.

..... [1]

(ii) Suggest a suitable reagent for reaction 2.

..... [1]

(iii) Reaction 3 is an elimination reaction.

Write an equation for this reaction and identify the solvent and conditions used.

equation .....

solvent and conditions .....
[2]

(iv) $C_2H_5CH_2OH$ can be directly converted to C_3H_6.

Suggest the reagent and conditions for this conversion.

..... [1]

3 The alkanes are a homologous series of organic molecules. Alkanes are generally unreactive and are commonly used as fuels.

(a) Define homologous series.

.....

..... [2]

(b) Give two reasons to explain the general unreactivity of alkanes.

1 .....

2 .....
[2]

(c) Alkanes with low relative molecular mass, M_r, are more useful than those found in heavier crude oil fractions.

Name the process that is used to obtain alkanes with low M_r from heavier crude oil fractions.

..... [1]

(e) Hexane can be converted into compounds **E** and **F** at high temperature and pressure. Fig. 3.2 shows the reaction scheme involving hexane, **E** and **F**.

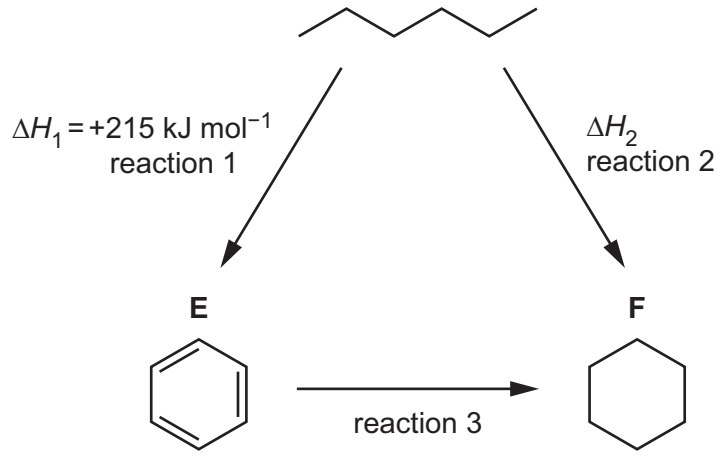


Fig. 3.2

(i) Identify a suitable reagent for reaction 3.

..... [1]

(iii) Write an equation for the complete combustion of hexane.

..... [1]

4 Fig. 4.1 shows how propane, C_3H_8, can be converted to propanoic acid, CH_3CH_2COOH.

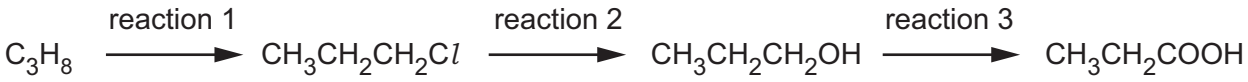


Fig. 4.1

(a) Reaction 1 in Fig. 4.1 takes place in the presence of sunlight.

$$C_3H_8 + Cl_2 \rightarrow CH_3CH_2CH_2Cl + HCl$$

The reaction takes place via initiation, propagation and termination steps.

(i) Name the mechanism shown by reaction 1.

..... [1]

(ii) Complete the mechanism for reaction 1.

Construct equations to describe the steps of the mechanism.

initiation $Cl_2 \rightarrow 2Cl\bullet$

propagation 1 .....

propagation 2 .....

termination ..... $\rightarrow CH_3CH_2CH_2Cl$ [3]

3 Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

(a) Define empirical formula.

.....

..... [1]

(c) Cyclopentane reacts with Cl_2 in the presence of ultraviolet light to form C_5H_9Cl.

(i) The reaction is initiated by the bond fission of Cl_2.

State the type of bond fission shown in the initiation step.

..... [1]

(ii) Complete the equations to show the two propagation steps that follow the initiation step.

propagation 1 C_5H_{10} + ... → C_5H_9• + ...

propagation 2 C_5H_9• + ... → .....

[2]

(iii) The final step is shown.

$$C_5H_9\bullet + Cl\bullet \rightarrow C_5H_9Cl$$

Give the name for this step in the reaction.

..... [1]

(d) Fig. 3.2 shows a reaction cycle involving cyclopentane, cyclopentene and C_5H_9Cl.

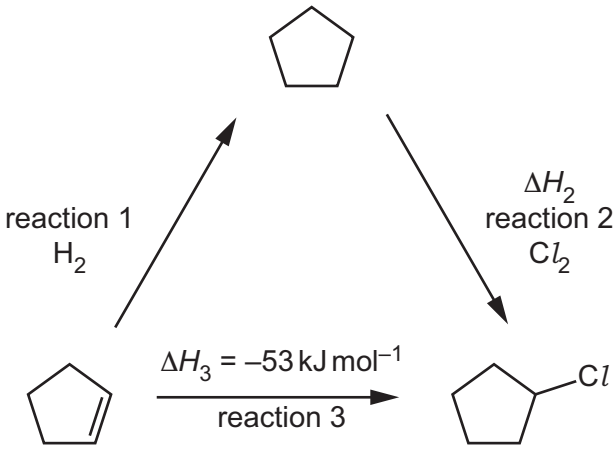


Fig. 3.2

(i) Identify a suitable reagent for reaction 3.

..... [1]

(e) Cyclopentene, C_5H_8, reacts with hot concentrated acidified $KMnO_4$ to form compound **W**, $C_5H_8O_4$.

(i) Draw the structure of **W**.

[1]

Table 3.3

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers) / cm^{-1}
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

[Total: 13]

4 Fig. 4.1 shows a possible synthesis of propene, C_3H_6.

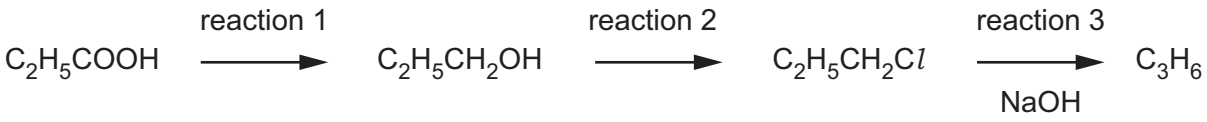


Fig. 4.1

(a) (i) Identify the type of reaction that occurs in reaction 1.

..... [1]

(ii) Suggest a suitable reagent for reaction 2.

..... [1]

(iii) Reaction 3 is an elimination reaction.

Write an equation for this reaction and identify the solvent and conditions used.

equation .....

solvent and conditions .....
[2]

(iv) $C_2H_5CH_2OH$ can be directly converted to C_3H_6.

Suggest the reagent and conditions for this conversion.

..... [1]

5 The skeletal formula of **W** is shown in Fig. 5.1.

W

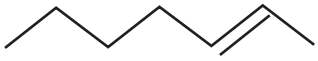


Fig. 5.1

(a) **W** has two positional isomers. Only one shows stereoisomerism.

Draw the structures of the **two** positional isomers of **W** in the boxes.

[2]

(d) **W** reacts with reagent **X** to produce ethanoic acid and pentanoic acid.

(i) State the role of **X** in this reaction.

..... [1]

(ii) Identify **X** and state the conditions used for this reaction.

.....

..... [1]

3 **(a)** Different hydrocarbon mixtures produced from fractional distillation of crude oil have different uses.

(i) State the compound that is heated with long-chain hydrocarbons to produce more useful smaller alkanes and alkenes.

..... [1]

(ii) Describe how photochemical smog is produced during the combustion of petrol in an internal combustion engine.

.....

.....

..... [2]

(b) C_4H_8 reacts with an excess of H_2 to produce C_4H_{10}.

reaction 1 $C_4H_8 + H_2 \rightarrow C_4H_{10}$

(i) Name a catalyst for reaction 1.

..... [1]

5 **(a)** **W** is a colourless liquid.

W

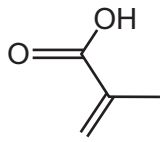


Fig. 5.1

(i) Deduce the empirical formula of **W**.

..... [1]

(ii) Two different reagents are each added to separate samples of **W** as shown in Table 5.1.

Complete Table 5.1.

Table 5.1

reagent and conditions	observation when reagent is added to **W**	structural formula of organic product when reagent is added to **W**
solid Na		
Br_2 in absence of ultraviolet light		

[4]

(c) Poly(ethene) is an addition polymer made from ethene.

Explain why ethene reacts with electrophiles but poly(ethene) does **not**.

.....

..... [1]

6 **(a)** HOC*l* reacts with $CH_2=CH_2$ to produce $HOCH_2CH_2Cl$ in an electrophilic addition reaction.

(i) Define addition reaction.

.....

..... [1]

(iii) State **one** use for HOC*l*.

.....

..... [1]

(b) Complete Fig. 6.1 to show the mechanism for the reaction between HOC*l* and $CH_2=CH_2$ to produce $HOCH_2CH_2Cl$.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



$HO^{\delta-}—Cl^{\delta+}$

Fig. 6.1

[4]

3 **(a)** C_3H_6 reacts with HBr(g) in an addition reaction.

(i) Define addition reaction.

.....

..... [1]

(ii) Complete Fig. 3.1 to show the mechanism for the addition reaction between C_3H_6 and HBr to produce 2-bromopropane. Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



Fig. 3.1

[4]

(iii) Explain why the major product of this reaction is 2-bromopropane rather than 1-bromopropane.

.....

.....

..... [2]

(e) Alkene **Z** contains two C=C bonds. **Z** reacts with an excess of hot concentrated acidified $KMnO_4$ to produce **only** CH_3COCH_3, $HOOCCH_2COOH$, CO_2 and H_2O.

Suggest the structure of **Z**.

[2]

3 **(a)** Hexene reacts with hydrogen gas to produce hexane.

reaction 1 $C_6H_{12}(l) + H_2(g) \rightarrow C_6H_{14}(l)$

Fig. 3.1 shows the distribution of energies of $H_2(g)$ molecules at temperature *T*.

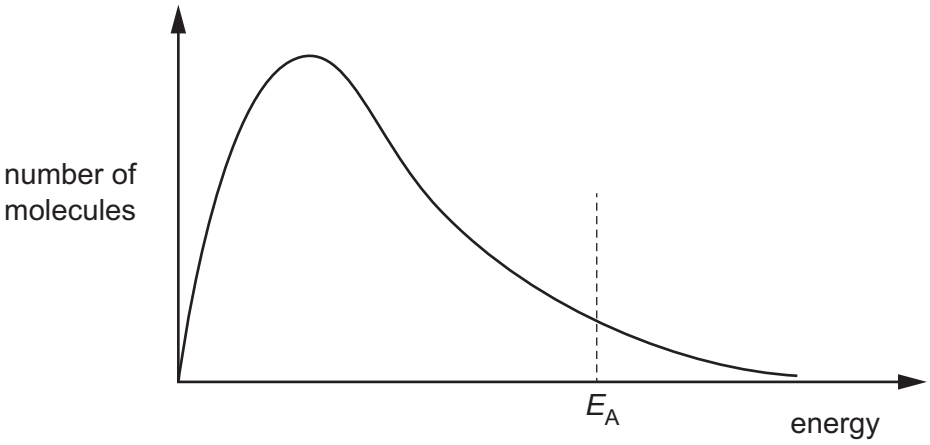


Fig. 3.1

(i) Sketch on Fig. 3.1 the shape of the curve for the same sample of $H_2(g)$ molecules when the temperature is increased. [2]

(iii) State the role of nickel when it is added to reaction 1.

..... [1]

4 The structure of vitamin C is shown in Fig. 4.1.

vitamin C

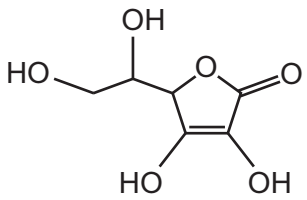


Fig. 4.1

(a) Deduce the empirical formula of vitamin C.

..... [1]

(b) The concentration of vitamin C is found by titration with $I_2(aq)$.

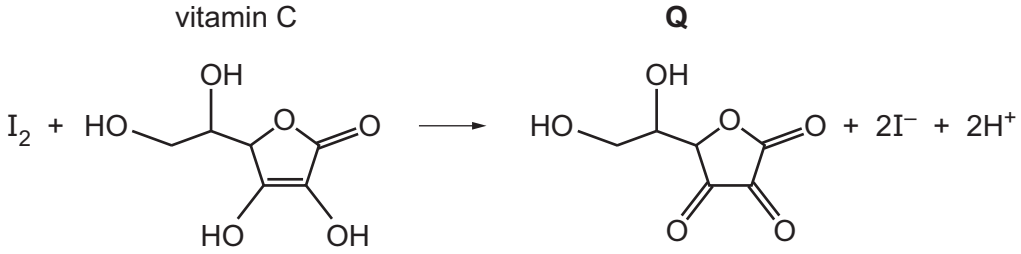


Fig. 4.2

A vitamin C tablet is dissolved in water to produce 200.0 cm^3 of vitamin C solution.
5.00 cm^3 of this vitamin C solution is added to a flask with approximately 150 cm^3 of water and an indicator.
Exactly 28.40 cm^3 of 5.00×10^{-4} mol dm^{-3} $I_2(aq)$ reacts with the sample of vitamin C solution in the flask.

[M_r: vitamin C, 176]

(i) Calculate the amount, in mol, of $I_2(aq)$ added to the flask in this titration.

amount of $I_2(aq)$ = ... mol [1]

(iv) Suggest **two** reasons why hot concentrated acidified potassium manganate(VII) is **not** a suitable reagent for producing **Q** from vitamin C.

1 .....

.....

2 .....

.....

[2]

1 Phosphorus and chlorine are elements in Period 3 of the Periodic Table.

(a) Chlorine forms three different compounds with phosphorus.

The most common compounds are PCl_3 and PCl_5.

(i) Complete Table 1.1.

Table 1.1

compound	oxidation number of P	oxidation number of Cl
PCl_3		
PCl_5		

[2]

(c) 1,2-dichloroethane, $ClCH_2CH_2Cl$, reacts with NaOH to produce an unsaturated compound, C_2H_3Cl, as shown in reaction 7.

reaction 7 $ClCH_2CH_2Cl + NaOH \rightarrow C_2H_3Cl + H_2O + NaCl$

(i) State what is meant by unsaturated.

.....

..... [1]

(ii) State the conditions for reaction 7.

..... [1]

3 The halogens chlorine, bromine and iodine show trends in chemical and physical properties down the group.

Table 3.1 shows some properties of chlorine, bromine and iodine.

Table 3.1

property	chlorine	bromine	iodine
colour and state at room temperature	green gas		
bond energy / kJ mol^{-1}	242	193	151
electronegativity	3.0	2.8	2.5
formula of sodium halide	NaCl	NaBr	NaI

(a) (i) Complete Table 3.1. [1]

(d) Iodine monobromide, IBr, is a dark red solid that melts near room temperature.

IBr reacts with propene. The mechanism for this reaction is the same as the mechanism for that of HBr with propene.

(i) Identify **all** the intermolecular forces that exist between molecules of IBr.

.....

..... [1]

(ii) Name the mechanism involved in the reaction of IBr with propene.

..... [1]

(iii) The reaction of IBr with propene forms two structural isomers.

Draw the **two** structural isomers shown by these molecules.

[2]

(v) Explain why the two structural isomers do **not** form in equal amounts.

.....

.....

..... [2]

1 Cobalt, rhodium and iridium are metals in the same group of the Periodic Table.

(a) The shorthand electronic configuration of cobalt is [Ar]$3d^7 4s^2$.

(i) Identify what is meant by [Ar] by giving its full electronic configuration.

..... [1]

(d) Hydrated rhodium(III) chloride, $RhCl_3 \cdot xH_2O$, catalyses the conversion of ethene to but-2-ene.

Both stereoisomers of but-2-ene are formed in the reaction.

(i) Hydrated rhodium(III) chloride contains 20.5% by mass of water of crystallisation.

Deduce the integer value of x in $RhCl_3 \cdot xH_2O$.

Show your working.

x = ...
[2]

(iii) Explain how the conversion of ethene to but-2-ene can be described as an addition reaction.

.....

..... [1]

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2 Chlorine is one of the elements in Group 17 of the Periodic Table.

(a) **(i)** Describe the colours of the Group 17 elements, chlorine to iodine, at room temperature.

.....

..... [1]

(ii) Describe the relative reactivity of the elements chlorine to iodine as oxidising agents.

..... [1]

(iii) State what is observed when chlorine reacts with hydrogen.

..... [1]

(iv) Explain why the thermal stability of the hydrogen halides decreases down the group.

.....

..... [1]

(b) The halogenoalkane CH_3CH_2Cl forms when chlorine reacts with C_2H_6 via a free-radical substitution mechanism.

(i) Define free radical.

.....

..... [1]

2 Oxygen is a Group 16 element.

(a) **(i)** Write equations for the following reactions.

- sodium and oxygen

.....

- sulfur and oxygen

.....

[2]

(iii) Draw the structures of the two alcohols formed in the reaction shown in equation 1.

equation 1 $C_3H_6(g) + H_2O(g) \xrightarrow{H_3PO_4} C_3H_8O(g)$

[2]

4 Bromoalkanes are used widely in industry, although there is increasing concern about their environmental impact.

Fig. 4.1 shows a reaction scheme involving 1,2-dibromoethane.

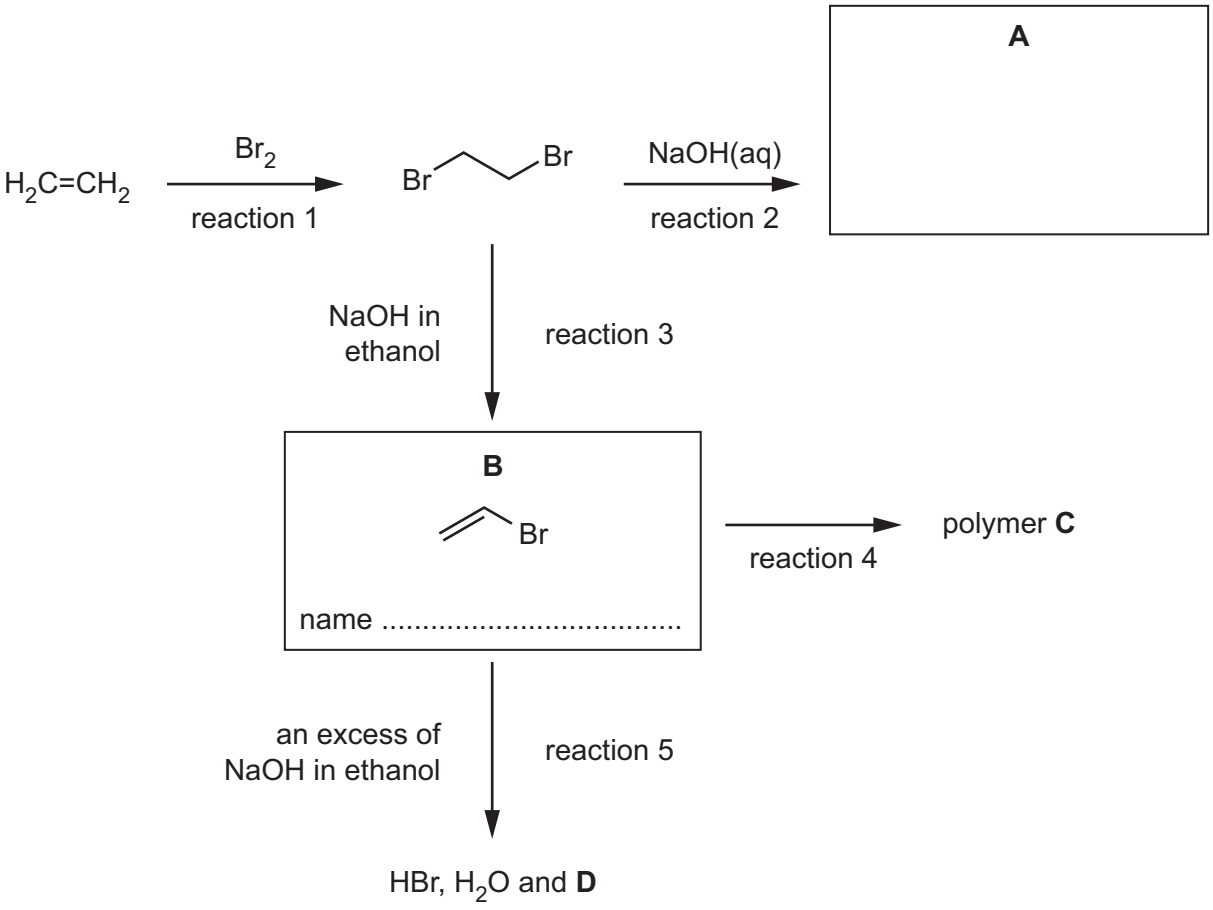


Fig. 4.1

(a) Complete Fig. 4.2 to show the mechanism for the formation of 1,2-dibromoethane in reaction 1.

Include charges, dipoles, lone pairs of electrons and curly arrows as appropriate.



Fig. 4.2

[3]

1 Cobalt, rhodium and iridium are metals in the same group of the Periodic Table.

(a) The shorthand electronic configuration of cobalt is [Ar]$3d^7 4s^2$.

(i) Identify what is meant by [Ar] by giving its full electronic configuration.

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Both stereoisomers of but-2-ene are formed in the reaction.

(i) Hydrated rhodium(III) chloride contains 20.5% by mass of water of crystallisation.

Deduce the integer value of x in $RhCl_3 \cdot xH_2O$.

Show your working.

x = ...
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.....

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BLANK PAGE

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(a) (i) Describe the colours of the Group 17 elements, chlorine to iodine, at room temperature.

.....

..... [1]

(iii) State what is observed when chlorine reacts with hydrogen.

..... [1]

(ii) State the essential condition for chlorine to react with C_2H_6 at room temperature.

..... [1]

(iii) Write **two** equations to show the propagation steps in this reaction.

1 .....

2 .....
[2]

5 **(a)** **M** reacts to form **R** by the addition of one reagent, as shown in Fig. 5.1.

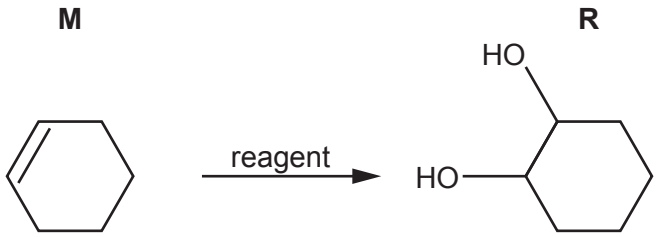


Fig. 5.1

Identify the reagent and conditions for this reaction.

..... [1]

(b) **R** is also made from **M** by two steps, as shown in Fig. 5.2.

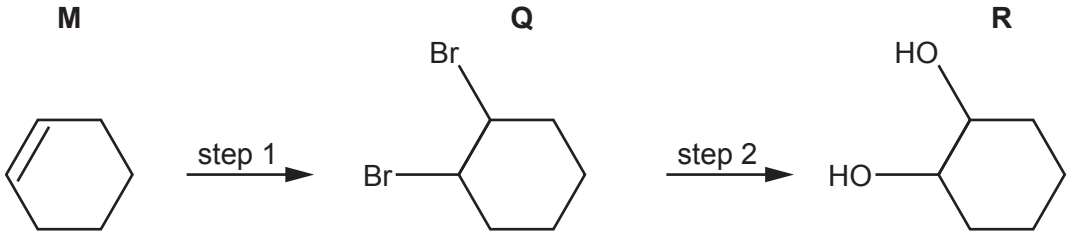


Fig. 5.2

(i) Identify the reagents and conditions for steps 1 and 2 in Fig. 5.2.

step 1 .....

step 2 .....

[2]

(ii) Name the mechanism for step 1 in Fig. 5.2.

..... [1]

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4 $CH_3(CH_2)_5CHBrCH_3$ exists as a pair of stereoisomers.

(a) Draw the three-dimensional structures of the **two** stereoisomers of $CH_3(CH_2)_5CHBrCH_3$. R can be used to represent $CH_3(CH_2)_5$.

[2]

(b) A sample of $CH_3(CH_2)_5CHBrCH_3$ reacts with NaOH to make $CH_3(CH_2)_5CH(OH)CH_3$ in an S_N1 mechanism.

Complete Fig. 4.1 to show the mechanism for the reaction of $CH_3(CH_2)_5CHBrCH_3$ and NaOH.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



Fig. 4.1

[3]

(c) Separate samples of $CH_3(CH_2)_5CHBrCH_3$, $CH_3(CH_2)_5CH(OH)CH_3$ and $CH_3(CH_2)_5CHCH_2$ are tested with different reagents.

Complete Table 4.1. If no reaction occurs, write × in the relevant box.

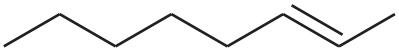
Table 4.1

reagent added	observation with $CH_3(CH_2)_5CHBrCH_3$	observation with $CH_3(CH_2)_5CH(OH)CH_3$	observation with $CH_3(CH_2)_5CHCH_2$
Br_2(l) in the dark			
PCl_5(s)			
$AgNO_3$(aq)			

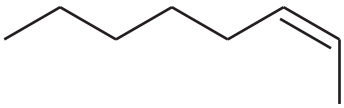
[3]

(d) $CH_3(CH_2)_5CHBrCH_3$ is heated with **D** to produce three different molecules, **E**, **F** and **G**.

E



F



G

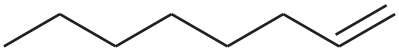


Fig. 4.2

(i) Name the type of reaction.

..... [1]

(ii) Identify **D** and the conditions used.

..... [1]

5 Hydrocarbon molecules contain covalent bonds.

(a) Define covalent bond.

.....

..... [1]

(c) In electrophilic reactions involving alkenes the π bond of C=C is broken.

(i) Suggest **one** difference between σ and π bonds that explains why the π bond is broken in electrophilic addition reactions involving alkenes.

.....

..... [1]

(ii) Complete Fig. 5.1 to show the mechanism for the electrophilic addition of hydrogen bromide to 2-methylpropene to produce the major organic product.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



Fig. 5.1

[4]

1 Bismuth is an element in Group 15 of the Periodic Table.

(a) Bismuth has metallic bonding.

Draw a labelled diagram to show the metallic bonding in bismuth.

[1]

(d) $NaBiO_3$ is an oxidising agent with similar properties to $KMnO_4$.

Fig. 1.1 shows an example of the use of $NaBiO_3$ as an oxidising agent.

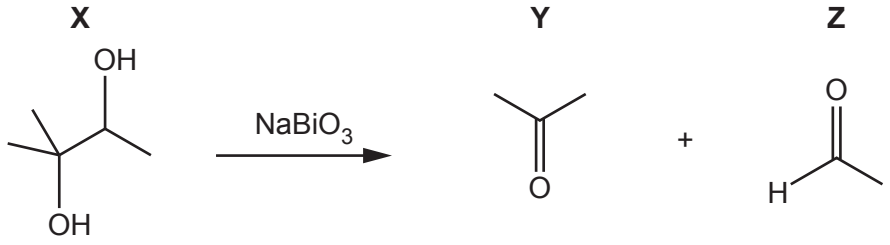


Fig. 1.1

(i) Explain the term oxidising agent.

.....

..... [1]

(ii) Compound **X** forms when methylbut-2-ene reacts with $KMnO_4$.

State the essential conditions for this reaction.

..... [1]

2 Chlorine, Cl_2, reacts with many elements and compounds to form chlorides.

Table 2.1 shows information about some chlorides of Period 3 elements.

Table 2.1

	Na	Mg	Si
formula of chloride			
structure of chloride	giant		
bonding of chloride			covalent
pH of solution formed on addition of chloride to water		6.2	

(a) Complete Table 2.1. [3]

(d) Cl_2 reacts readily with propene to form **K**, 1,2-dichloropropane.

K can be used to form **L**.

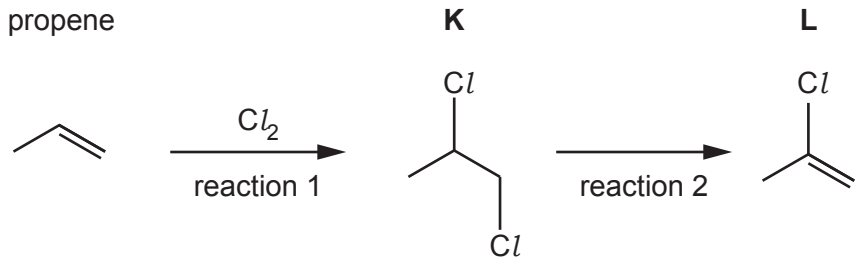


Fig. 2.1

(i) Complete Fig. 2.2 to show the mechanism for the reaction of Cl_2 with propene in reaction 1.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

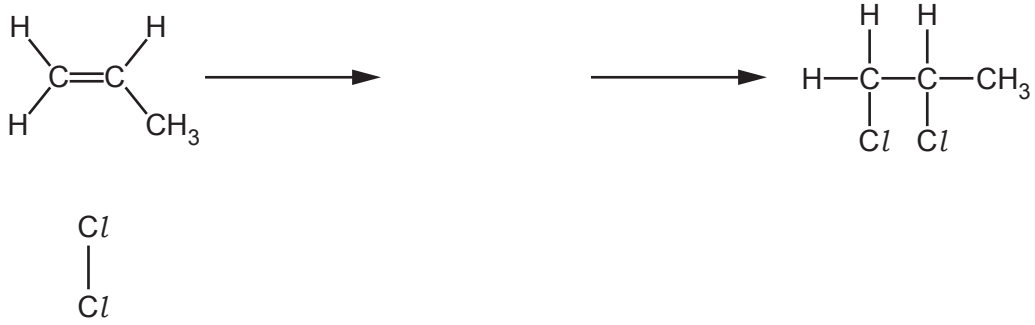


Fig. 2.2 [4]

(ii) Identify the reagent and conditions for reaction 2.

..... [1]

4 Compounds **C** and **D** are alkenes with the same molecular formula, C_5H_{10}.



Fig. 4.1

(a) (i) Give the systematic name of **D**.

..... [1]

(iv) Give the structural formula of the compound formed when **D** reacts with $H_2(g)$ in the presence of a Pt catalyst.

..... [1]

(v) **C** can form an addition polymer.

Draw the structure of **one** repeat unit of this addition polymer.

[1]

(c) **C** and **D** both react with HBr.

(i) **C** reacts with HBr to form **E**.

Complete the diagram in Fig. 4.2 to show the mechanism for this reaction.

Draw the structure of the organic intermediate.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



Fig. 4.2

[3]

(ii) **D** reacts with HBr to produce **F**, a chiral bromoalkane.

Draw the structure of **F**.

F

[1]

(iii) Explain why the reaction of HBr with **C** and **D** produces different major products.

C **D**



Fig. 4.3

.....

.....

.....

..... [2]

3 Phosphoric(V) acid, H_3PO_4, is used in both inorganic and organic reactions.

(a) H_3PO_4 is made in a two-step process from phosphorus.

step 1 Phosphorus reacts with an excess of oxygen to form a white solid.

step 2 The white solid then reacts with water to form H_3PO_4.

(i) Write an equation for each step.

step 1 .....

step 2 .....

[2]

(c) Fig. 3.1 shows a reaction scheme that involves H_3PO_4 in several reactions.

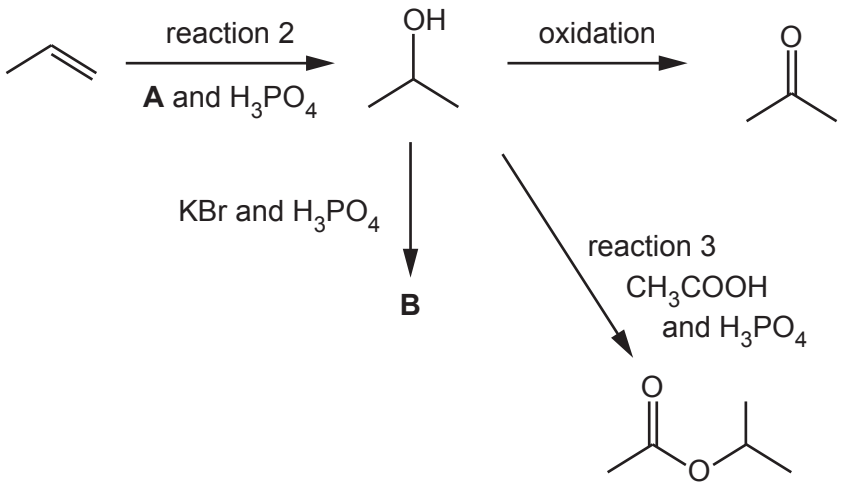


Fig. 3.1

(i) Identify **A**, which reacts with propene in the presence of H_3PO_4 in reaction 2.

..... [1]

4 Lactic acid, $CH_3CH(OH)COOH$, and pyruvic acid, $CH_3COCOOH$, both contain two functional groups.

lactic acid **pyruvic acid**

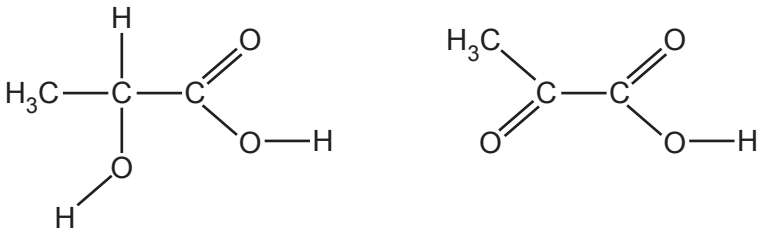


Fig. 4.1

(a) (i) Explain why lactic acid exists as optical isomers.

.....

.....

..... [1]

(b) Two possible syntheses of pyruvic acid are shown in Fig. 4.3 and Fig. 4.4.

Each synthesis has a total of three steps.

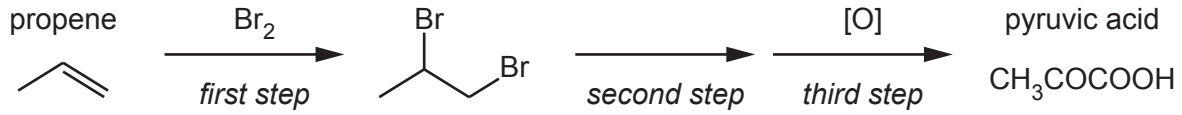


Fig. 4.3

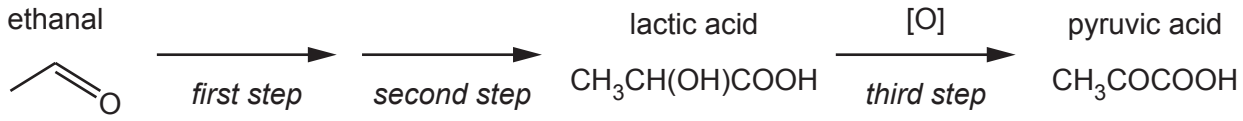


Fig. 4.4

(i) Complete the diagram in Fig. 4.5 to show the mechanism for the reaction of propene with Br_2.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



Fig. 4.5

[3]

4 Compounds **C** and **D** are alkenes with the same molecular formula, C_5H_{10}.



Fig. 4.1

(a) (i) Give the systematic name of **D**.

..... [1]

(iv) Give the structural formula of the compound formed when **D** reacts with $H_2(g)$ in the presence of a Pt catalyst.

..... [1]

(v) **C** can form an addition polymer.

Draw the structure of **one** repeat unit of this addition polymer.

[1]

(c) **C** and **D** both react with HBr.

(i) **C** reacts with HBr to form **E**.

Complete the diagram in Fig. 4.2 to show the mechanism for this reaction.

Draw the structure of the organic intermediate.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



Fig. 4.2

[3]

(ii) **D** reacts with HBr to produce **F**, a chiral bromoalkane.

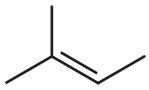
Draw the structure of **F**.

F

[1]

(iii) Explain why the reaction of HBr with **C** and **D** produces different major products.

C



D

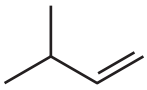


Fig. 4.3

.....

.....

.....

..... [2]

6 Compound **W**, CH_2=CHCN, is used to make an addition polymer which is present in carbon fibres.

(a) Draw **one** repeat unit of the addition polymer of **W**.

[1]

(b) CH_3CHO is used in a two-step synthetic route to form **W**, as shown in Fig. 6.1.
In step 1, CH_3CHO is heated with HCN in the presence of KCN.

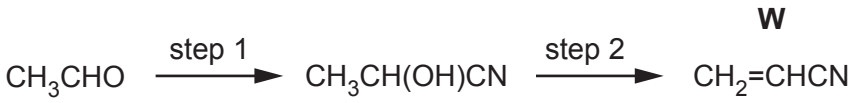


Fig. 6.1

(i) Name the mechanism for the reaction in step 1 in Fig. 6.1.

..... [1]

(iii) Suggest a suitable reagent and conditions for step 2 in Fig. 6.1.

..... [1]

4 In industry, ethanol is made by reacting ethene with steam in the presence of H_3PO_4.

reaction 1 $C_2H_4(g) + H_2O(g) \rightleftharpoons C_2H_5OH(g)$

(a) Use the bond energy values in Table 4.1 to calculate the enthalpy change, ΔH_r, for reaction 1.

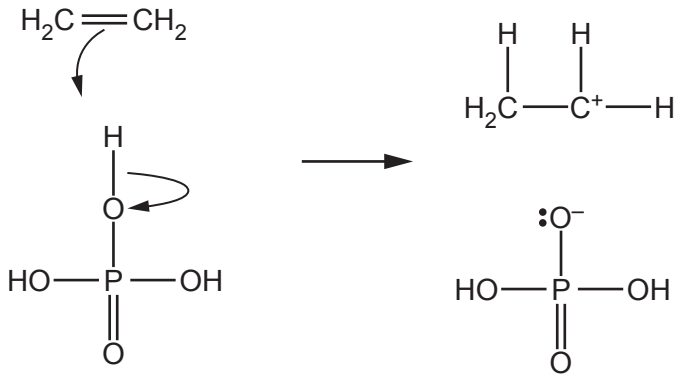
Table 4.1

bond	bond energy / kJ mol^{-1}
C–C	350
C=C	610
C≡C	840
C–H	410
C–O	360
C=O	740
O–H	460

ΔH_r = ... kJ mol^{-1} [2]

(c) The mechanism for reaction 1 can be described in three steps. Steps 1 and 2 for reaction 1 are shown in Fig. 4.1.

step 1



step 2

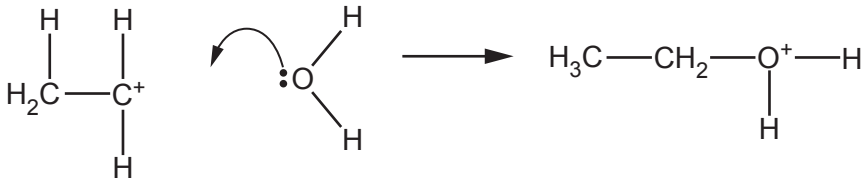


Fig. 4.1

(i) Describe the behaviour of H_3PO_4 in step 1 in Fig. 4.1. Explain your answer.

.....

..... [1]

(iii) Complete Fig. 4.2 to show the mechanism for step 3 of reaction 1. Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

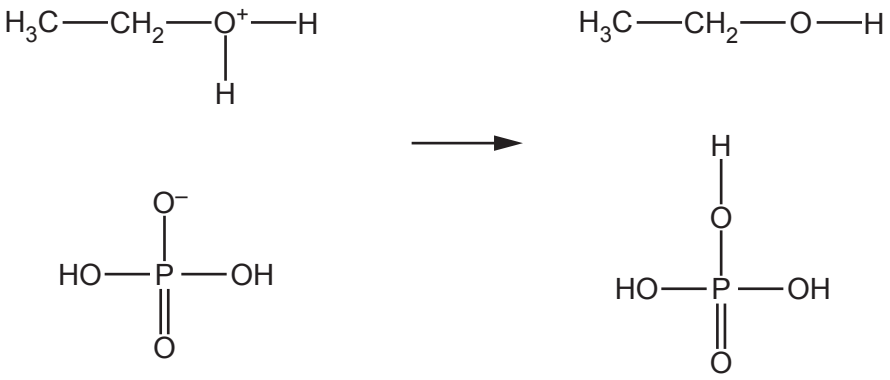


Fig. 4.2

[2]

(vi) Propene also reacts with steam. A mixture of organic products is produced.

Explain why propan-2-ol is produced in the higher yield.

.....

.....

..... [2]

5 **(a)** Describe structural isomerism.

.....

..... [1]

(b) **A** and **B** are structural isomers with molecular formula $C_5H_{10}O$.

They are both straight-chained molecules with only one functional group.

Table 5.1 describes observations when separate samples of **A** and **B** are added to different reagents.

Table 5.1

reagent	**A**	**B**
2,4-dinitrophenylhydrazine (2,4-DNPH reagent)	orange precipitate appears	orange precipitate appears
Tollens' reagent	silver mirror appears	no reaction
alkaline I_2(aq)	no reaction	no reaction

(i) Name the functional group present in both **A** and **B**.

..... [1]

(c) **C** is a structural isomer of **A** and **B**.

C is straight chained and has two functional groups.

C shows only one type of stereoisomerism.

Table 5.2 describes observations when separate samples of **C** are added to different reagents.

Table 5.2

reagent	**C**
2,4-dinitrophenylhydrazine (2,4-DNPH reagent)	no reaction
$Br_2(aq)$	orange to colourless
alkaline $I_2(aq)$	yellow precipitate appears

(i) Draw the structure of **C** in the box.

C

[2]

3 Chlorine is a very reactive element.

(a) Chlorine reacts with silicon to form silicon(IV) chloride.
Describe the appearance of silicon(IV) chloride at room temperature and pressure. State its structure and bonding.

appearance .....

structure and bonding .....
[2]

(d) Chlorine reacts with methane in a series of reactions to produce chloroalkanes.

(i) State the conditions required for chlorine to react with methane.

..... [1]

(ii) One of the products of the reaction is CH_2Cl_2 which reacts further to produce $CHCl_3$.

Complete Table 3.2 to show details of the mechanism that forms $CHCl_3$ from CH_2Cl_2.

Table 3.2

name of step	equation
initiation	.....
propagation	$CH_2Cl_2 + Cl\bullet \rightarrow$ .....
termination	..... $\rightarrow CHCl_3$

[3]

4 **V** is a colourless liquid.

V

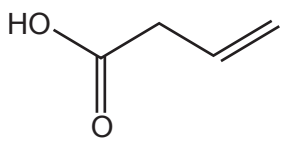


Fig. 4.1

(a) **V** reacts with an excess of $LiAlH_4$ to form **W**.

(i) Draw the structure of **W** in the box.

W

[1]

(b) **V** reacts to form **Z** in a single reaction, as shown in Fig. 4.2.

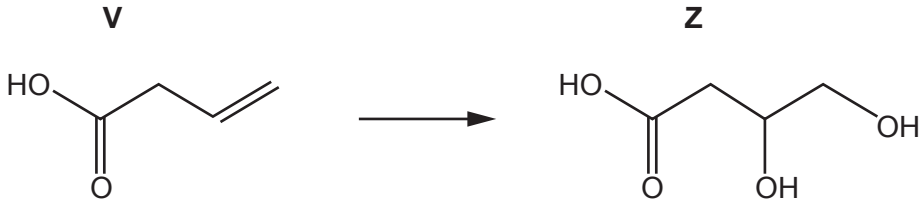


Fig. 4.2

(i) Suggest the reagent and conditions needed to form **Z** from **V**.

..... [1]

5 **(a)** Molecule **M** is present in petrol, a fuel used in cars. **M** is a saturated, non-cyclic hydrocarbon. **M** contains eight carbon atoms.

(i) Construct an equation for the complete combustion of **M**.

..... [2]

(ii) Describe how the composition of products differs when incomplete combustion of **M** occurs.

.....

..... [2]

3 Alkenes undergo an addition reaction with a 1:1 mixture of CO and H_2 to form aldehydes.

Fig. 3.1 shows the reaction of propene with a 1:1 mixture of CO and H_2.

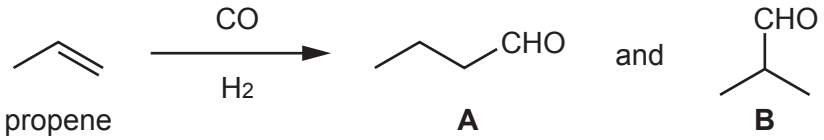


Fig. 3.1

(a) (i) Define addition reaction.

.....

..... [1]

(iv) The complete reaction of propene with a 1:1 mixture of CO and H_2 produces **A** and **B** only. The product mixture contains 96% **A** and 4% **B**.

Calculate the mass of **A** produced in this reaction when 5.00×10^3 kg of propene is used.

mass of **A** = ... kg [1]

(ii) Table 3.1 shows information about three experiments involving **B**.

Complete Table 3.1.

Table 3.1

experiment	reagents	observation with **B**
1		solution turns from orange to green
2		a silver mirror forms on the sides of the reaction vessel
3	Br_2(aq)	

[3]

(c) When propene reacts with CO and an excess of H_2, an alkane and a mixture of alcohols are formed instead. The alcohols are isomers of each other.

Suggest the molecular formulae of the alkane and the alcohols that are formed under these conditions.

molecular formula of alkane .....

molecular formula of alcohols .....

[2]

4 Fig. 4.1 shows some reactions of compound **D**, 2-bromobutane.

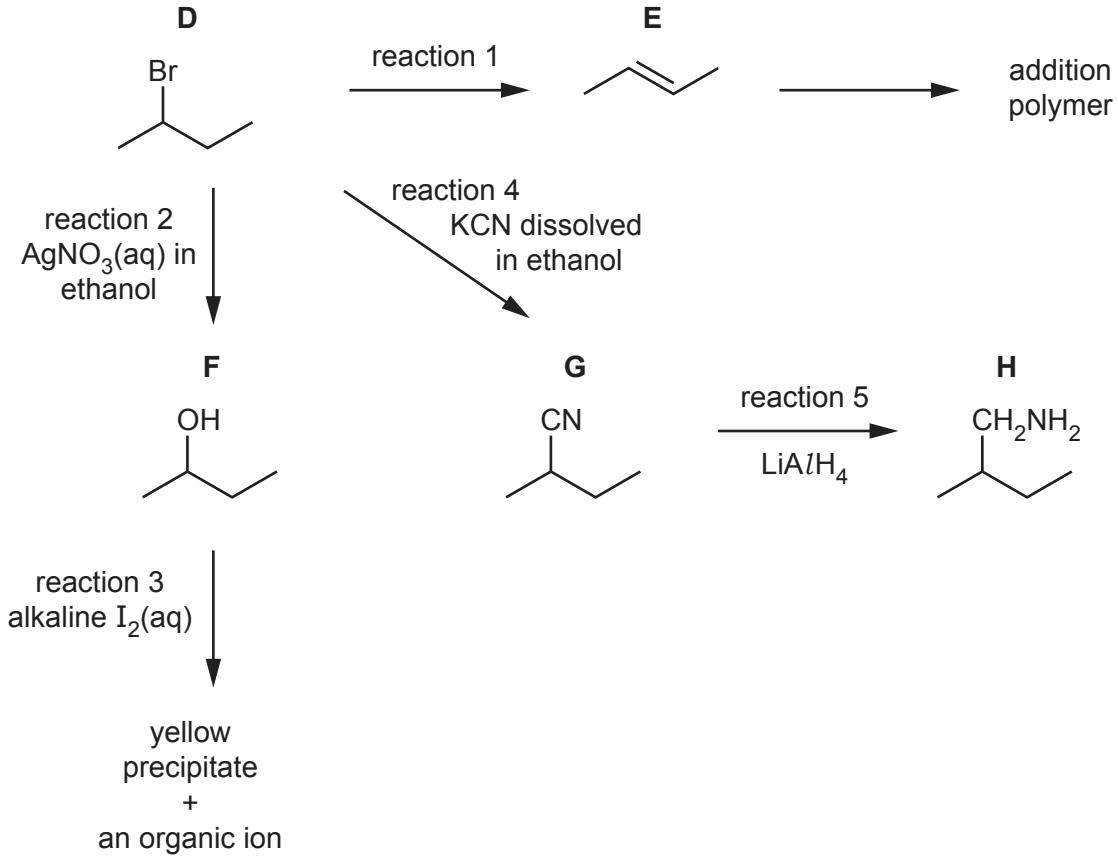


Fig. 4.1

(a) (i) State the reagent and conditions used to form **E** in reaction 1.

..... [1]

(iii) **E** also forms when **F** is heated strongly in the presence of an Al_2O_3 catalyst.

Write an equation for this reaction.

..... [1]